

Assignment No. - 1

ON

DevOps And Agile Foundation - SSCS4010

TITLE: Git & GitHub Basics (Clone, Commit, Branches)

BACHLEOR OF SCIENCE IN INFORMATION TECHNOLOGY (Honr.)

SUBMITTED TO:

Name: Mr. Kaushal Singh V. and Mr. Rishi Kumar Sharma

Designation: Assistant professor and Lecturer

Lab Assignment Title: Git & GitHub Basics (Clone, Commit, Branches)

P P Savani University

Max. Marks: 50

Marks Obtained: _____

Faculty Signature: _____

SUBMITTED BY:

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Class: BSCIT-H – 7A - Batch 2023-27

INSTITUTE OF COMPUTER SCIENCE AND APPLICATIONS

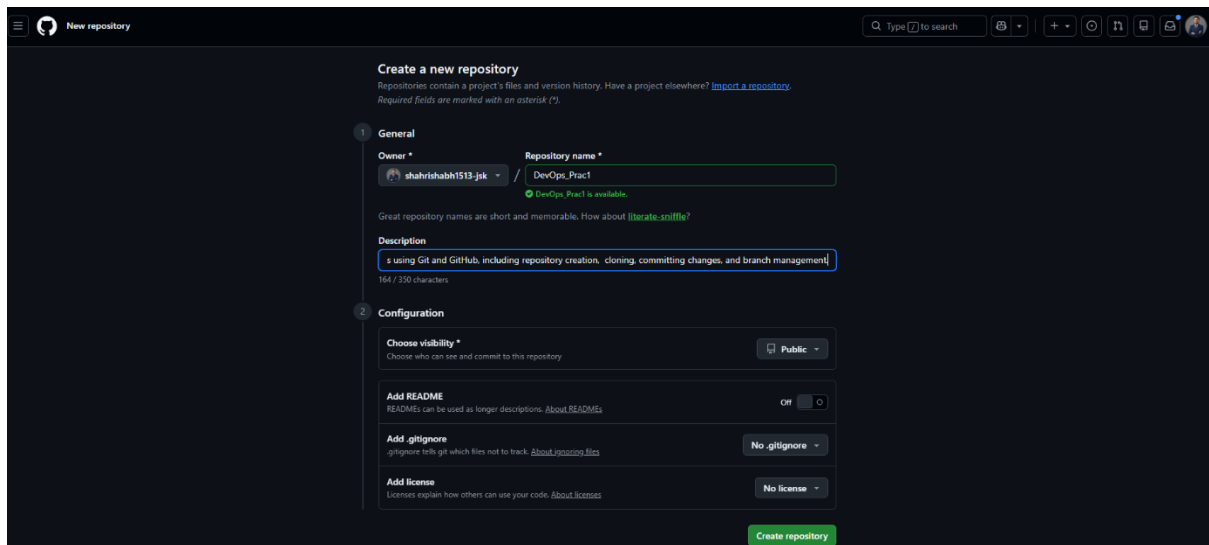
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MANGROL, SURAT (GUJARAT)

1. Verified the successful installation of Git by executing the version command and confirming that Git was installed correctly on the system.

```
C:\Users\len21>git --version
git version 2.52.0.windows.1
```

2. Created a new public GitHub repository named **DevOps_Prac1** by entering the repository details and configuring the required settings before creation.



The screenshot shows the GitHub 'Create a new repository' interface. The 'General' section includes fields for 'Owner' (shahrishabh1513-jsk) and 'Repository name' (DevOps_Prac1). A description is provided: 's using Git and GitHub, including repository creation, cloning, committing changes, and branch management'. The 'Configuration' section shows 'Choose visibility' set to 'Public', 'Add README' set to 'Off', 'Add .gitignore' set to 'No .gitignore', and 'Add license' set to 'No license'. A 'Create repository' button is at the bottom right.

Repository URL: https://github.com/shahrishabh1513-jsk/DevOps_Prac1

3. Created a new README file and initialized a local Git repository successfully using the git init command.

```
PS D:\Devops_sem_7> echo "# DevOps_Prac1" >> README.md
PS D:\Devops_sem_7> git init
Initialized empty Git repository in D:/Devops_sem_7/.git/
```

4. Added all project files into the staging area and created the first commit to save the initial project version.

```
PS D:\Devops_sem_7> git add *
PS D:\Devops_sem_7> git commit -m "first commit"
[master (root-commit) 091b4d8] first commit
2 files changed, 8 insertions(+)
create mode 100644 README.md
create mode 100644 index.html
```

5. Changed the default branch name from master to main for better repository management and standard naming.

```
PS D:\Devops_sem_7> git branch -M main
```

6. Connected the local Git repository with the remote GitHub repository using the repository URL.

```
PS D:\Devops_sem_7> git remote add origin https://github.com/shahrishabh1513-jsk/DevOps_Prac1.git
```

7. Uploaded all committed project files from the local system to the GitHub remote repository successfully.

```
PS D:\Devops_sem_7> git push -u origin main
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Delta compression using up to 4 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (4/4), 364 bytes | 182.00 KiB/s, done.
Total 4 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/shahrishabh1513-jsk/DevOps_Prac1.git
 * [new branch]      main -> main
branch 'main' set up to track 'origin/main'.
```

8. Displayed the available branches and verified that the current working branch was set to main.

```
PS D:\Devops_sem_7> git branch
* main
```

9. Created a new branch named branch-2 to perform development separately from the main branch.

```
PS D:\Devops_sem_7> git branch branch-2
PS D:\Devops_sem_7> git branch
branch-2
* main
```

10. Switched the current working environment from the main branch to branch-2 successfully.

```
PS D:\Devops_sem_7> git switch branch-2
Switched to branch 'branch-2'
```

11. Created an additional branch named branch-3 for testing branch creation and branch management operations.

```
PS D:\Devops_sem_7> git branch branch-3
PS D:\Devops_sem_7> git branch
* branch-2
  branch-3
  main
```

12. Changed the active branch from branch-2 to branch-3 to continue branch-related operations.

```
PS D:\Devops_sem_7> git switch branch-3
Switched to branch 'branch-3'
```

13. Executed the merge command to combine changes and verified that all branches were already synchronized.

```
PS D:\Devops_sem_7> git merge main
Already up to date.
```

14. Returned to the main branch and completed the merge process to maintain the updated project version.

```
PS D:\Devops_sem_7> git switch main
Switched to branch 'main'
Your branch is up to date with 'origin/main'.
PS D:\Devops_sem_7> git merge branch-3
Already up to date.
```

15. Deleted the unnecessary branch after completing branch operations and keeping the repository organized.

```
PS D:\Devops_sem_7> git branch -d branch-2
Deleted branch branch-2 (was 091b4d8).
```